

JAN HUERTAS MARTIN

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My research investigates the **physical mechanisms that control chromatin accessibility**. Using **multiscale molecular simulations** integrated with experimental data, I study how nucleosome dynamics, chromatin organization, and protein-protein interactions shape transcription factor search and activity in chromatin. A particular focus of my work is **understanding how pioneer transcription factors access nucleosome-occupied DNA** and how these mechanisms can be harnessed to improve **cellular reprogramming**.

ACADEMIC APPOINTMENTS

Independent Research Fellow – Herchel Smith Fellowship University of Cambridge	2023-present
Junior Research Fellow St Catharine's College	2023-present
Postdoctoral Fellow University of Cambridge	2021-2024
HORIZON TMA Marie Skłodowska-Curie Action Postdoctoral Fellow	2022-2024

EDUCATION

Dr.rer.Nat. in Chemistry (Theoretical Chemistry) Max Planck Institute for Molecular Biomedicine - Westfälische Wilhelms Universität Münster (WWU). CiM-IMPRS Graduate Program.	2016-2021
M.Sc. in Bioinformatics for Health Sciences Universitat Pompeu Fabra, Barcelona	2014-2016
B.Sc. in Human Biology Universitat Pompeu Fabra, Barcelona	2010-2014

SELECTED PUBLICATIONS

★ *First/Co-First author.* © *Computational Modelling Lead.* § *Corresponding/Co-corresponding author*

- ★ Huertas J, Maristany MJ, Colleparado-Guevara R. **Oct4 clusters promote DNA accessibility by enhancing chromatin plasticity.** *BioRxiv*. (2026)
- © Zhou H, Huertas J, [...] Colleparado-Guevara R, Rosen MK. **Multi-scale structure of chromatin condensates rationalizes phase separation and material properties.** *Science* (2025)
- ★ Huertas J and MacCarthy CM, [...] Schöler HR, Cojocaru V. **OCT4 interprets and enhances nucleosome flexibility.** *Nucleic Acids Research* (2022)
- § Perez-Lopez JI and Maristany MJ, Farr SE, Huertas J* and Colleparado-Guevara R*. **A Goldilocks zone of DNA flexibility defines stable yet plastic nucleosomes, tuned by histone chemistry.** *Nature Communications* - in Press (2026)
- © Shimazoe MA, Huertas J, [...] Colleparado-Guevara R and Maeshima K. **Linker histone H1 functions as a liquid-like glue to organize chromatin in living human cells.** *Science Advances* (2026)

FULL LIST OF PUBLICATIONS

6. Zhang K, Maristany MJ, **Huertas J**, Colleparado-Guevara R, Ramani V. **Single-molecule nucleosome spacing coordinates chromatin fiber interactions.** *BioRxiv* (2026)
7. ★ **Huertas J**, Huxford J, Maristany MJ, Colleparado-Guevara R, Gibson B. **Bind me if you can: Transcription factor target search in condensed chromatin.** *Under Review in Current Opinion in Structural Biology*
8. Chen Y, **Huertas J**, et al. **A Spectrum of Free Energy Landscape Topologies Encodes Chromatin Polymorphism and Phase Separation.** *BioRxiv* (2026)
9. Maristany MJ, Russell K, Perez-Lopez JI, **Huertas J**, Colleparado-Guevara R. **Molecular Simulations of Chromatin-Associated Biomolecular Condensates.** *Current Opinion in Structural Biology* (2026)
10. ★ **Huertas J**, Maristany MJ, Perez Lopez JI, Colleparado-Guevara R. **Pushing the experimental resolution boundary in chromatin fibre organisation using multiscale simulations.** *Current Opinion in Genetics and Development* (2026)
11. Maristany MJ, **Huertas J**, et al. **Multiscale simulations of chromatin structure and chromatin phase transitions.** *Springer Nature, Methods in Molecular Biology* – in Press (2026).
12. © Spicer MFD, Wijma S, Schütte N, **Huertas J** et al. **Electrostatic control of chromatin compaction safeguards against apoptotic DNA release.** *EMBO Journal* - in Press (2026)
13. Russel K, [...] **Huertas J**, Orozco M, Rosen MK, Colleparado-Guevara R. **Near-atomistic simulations reveal the molecular principles that control chromatin structure and phase separation.** *Nature Communications* - in Press (2026)
14. Li H and Dalglish JLT, Lister G, Maristany MJ, **Huertas J** et al. **Mapping chromatin structure at base-pair resolution unveils a unified model of cis-regulatory element interactions.** *Cell* (2025)
15. Bell C and Chen L and Maristany MJ, Blaukopt C, Zhou H, **Huertas J** et al. **An electrostatic repulsion model of centromere organization.** *BioRxiv* (2025)
16. Nakashima KN and Mihoubi FZ, [...] **Huertas J**, Rubio-Sánchez R, Colleparado-Guevara R, O'Flaherty DK, Bonfio C. **Differential stability and dynamics of DNA-based and RNA-based coacervates affect non-enzymatic RNA chemistry** *Nat. Comms* (2025)
17. Chen L, [...] **Huertas J**, Redding S, Colleparado-Guevara R, Rosen MK. **Nucleosome Spacing Can Fine-Tune Higher Order Chromatin Assembly.** *Nat. Comms* (2025)
18. © Montez M and Zhu D, **Huertas J**, [...] Colleparado-Guevara R, Dean C. **Cold-induced nucleosome dynamics linked to silencing of Arabidopsis FLC..** *Nat. Comms* (2025)
19. Maristany MJ, Aguirre A, Espinosa JR, **Huertas J**, Colleparado-Guevara R, Joseph JA. **Decoding Phase Separation of Prion-Like Domains through Data-Driven Scaling Laws.** *Elife* (2025)
20. Ng TLC and Hoare MP and Maristany MJ, Wilde EJ, Sneideris T, **Huertas J**, [...] Kumita JR. **Tandem-repeat proteins introduce tuneable properties to engineered biomolecular condensates.** *Chemical Science* (2024)
21. Orsetti A and Van Oosten D, [...] **Huertas J**, van Ingen H, Cojocar V. **Structural dynamics in chromatin unraveling by pioneer transcription factors.** *Biophysical Reviews* (2024)
22. ★ **Huertas J**, Woods E, Colleparado-Guevara R, **Multiscale modelling of chromatin organisation: Resolving nucleosomes at near-atomistic resolution inside genes.** *Current Opinion in Cell Biology.* (2022).
23. ★ **Huertas J**, Schöler HR, Cojocar V. **Histone tails cooperate to control the breathing of genomic nucleosomes.** *PLoS Computational Biology* (2021).
24. ★ **Huertas J**, Cojocar V. **Breaths, twists, and turns of atomistic nucleosomes.** *J. Mol. Biol.* (2021).
25. ★ **Huertas J**, MacCarthy CM, Schöler HR, Cojocar V. **Nucleosomal DNA Dynamics Mediate Oct4 Pioneer Factor Binding.** *Biophysical Journal* (2020).
26. © Viplav A., Saha J, **Huertas J**, [...] Galic M. **ArhGEF37 assists dynamin 2 during clathrin-mediated endocytosis.** *J. Cell Sci.* (2019).

GRANTS

FUNDING

- 2022 Herchel Smith Postdoctoral Research Fellowship (**from 06/2023 to 06/2026**)
- 2022 HORIZON Marie Skłodowska-Curie Action (MSCA) Postdoctoral Fellowship (**from 09/2022 to 06/2024**)

COMPUTING GRANTS

- 2024 EPSRC High Performance Computing Grant. (114,000 CUs)
- 2018 Project ID 12622 extension (20 million hours). Gauss Call 2/2018
- 2017 Computing grant from the GAUSS CENTER (20 million hours). Project ID 12622. Gauss Call 2/2017

SUPERVISION AND TEACHING

RESEARCH SUPERVISION

Co-supervisor of a PhD Student in the Collepardo lab.

Co-supervisor of M.Sc. and Part II students in the Collepardo lab.

(10 students)

Co-supervisor of visiting students in the Collepardo lab.

(7 students, including undergraduates and summer interns)

Co-supervisor of 2 M.Sc students in the Schöler lab.

TEACHING

Senior demonstrator Scientific Computing, Part IA (Department of Physics, University of Cambridge)

Lecturer Systems Biology, Part III (School of Biological Sciences, University of Cambridge)

Over 80 hours of small group teaching (Cambridge University "Supervisions").

(Mathematical Biology, Clare College and Emmanuel College, University of Cambridge)

Practical Courses, M.Sc. Program (Biology Department, University of Münster)

Lectures and practical courses, M.Sc. program (Chemistry Department, University of Münster)

TALKS AND POSTERS

TALKS

- 27/08/2026 **Multi-scale and multi-purpose simulations of DNA** (Lausanne, Switzerland) ***Invited Speaker**
- 15/06/2026 **ISQBP President meeting** (Cluj-Napoca, Romania) ***Invited Speaker**
- 30/05/2026 **Genome Organization and Dynamics in Gene Regulation – GRS** (Castelldefels, Spain)
- 25/02/2026 **2026 Annual Biophysical Society Meeting** (San Francisco, CA) ***Travel award**
- 15/02/2025 **2025 Annual Biophysical Society Meeting** (Los Angeles, CA) ***Session Chair**
- 27/11/2024 **Sociedad Argentina de Biofísica** (Bahía Blanca, Argentina) ***Invited Speaker**
- 24/06/2024 **International Union for Pure and Applied Biophysics 2024** (Kyoto, Japan)
- 19/06/2023 **Cambridge Epigenetics Club Summer Symposium** (Cambridge, UK) ***Best talk award**
- 09/05/2023 **4DGenomics2023: Investigating, modelling and understanding the genome in space and time** (Milano, Italy) ***Invited Speaker**
- 20/04/2023 **Multiscale simulations of DNA from electrons to nucleosomes: 22 years of the Ascona B-DNA Consortium** (Ascona, Switzerland)
- 20/02/2023 **2023 Biophysical Society Meeting** (San Diego, CA) ***Travel award / Poster prize**
- 16/11/2020 **Science for Life** (Utrecht – Virtual)
- 09/09/2020 **8th Annual German Stem Cell Network Conference** (Virtual)

21/07/2019 **Joint 12th EBSA 10th ICBP-IUPAP BIOPHYSICS CONGRESS** (Madrid)
26/04/2019 **Graduate School Meeting** (Münster, Germany)
16/11/2018 **Netherlands Society on Biomolecular Modelling meeting** (Utrecht)
21/06/2018 **ISQBP President meeting** (Barcelona)

POSTERS

01/06/2026 **Genome Organization and Dynamics in Gene Regulation – GRC** (Castelldefels, Spain)
20/05/2026 **Cellular mechanisms driven by phase separation – EMBL** (Heidelberg, Germany) ***Flash Talk**
23/02/2026 **2026 Annual Biophysical Society Meeting** (San Francisco, CA)
30/11/2025 **Sociedad Argentina de Biofísica** (Buenos Aires, Argentina)
19/02/2025 **2025 Annual Biophysical Society Meeting** (Los Angeles, CA)
29/11/2024 **Sociedad Argentina de Biofísica** (Bahía Blanca, Argentina)
12/02/2024 **2024 Annual Biophysical Society Meeting** (Philadelphia, PA) ***Travel award**
28/11/2023 **Sociedad Argentina de Biofísica** (Cordoba, Argentina)
04/04/2023 **5th Manchester Multiscale Conference** (Manchester, UK)
31/03/2023 **Physics of Emergent Behaviours IV Conference** (Harrogate, UK)
28/03/2023 **Physics of Life 2023** (Harrogate, UK)
01/04/2019 **Multiscale Modeling of Chromatin** (Les Houches, France) ***Poster Prize**
22/08/2018 **Biophysical Society Thematic Meeting** (Santa Cruz, CA)

OTHER RELEVANT EXPERIENCE

Member of the Chromatin Consortium MBL (2023-Ongoing)

Visiting scientist in the **Marine Biology Lab in Woods Hole, Massachussets**.

An international collaborative initiative among leading research groups in the study of chromatin, including the groups of Mike Rosen (UT Southwestern), Vijay Ramani (UCSF), Bryan Gibson (St Jude's Hospital), Daniel Gerlich (IMBA,Vienna), Sy Redding (UMass) and Rosana Collepardo-Guevara (University of Cambridge)

Organizing member of the following initiatives:

The Early-Career Symposia from the Center for Physical Biology (Cambridge, 2022-2025)

The Multiscale Genome Organization Talk Series (Online, 2024-2025)

The Young Biophysicists Group (YIB) (A subgroup of the Argentine Biophysics Society, 2023-2025)

The international CiM-IMPRS GRADUATE SCHOOL MEETING (Münster, 2019)

Various PhD student workshops on Scientific Computing, Statistics, and Bioinformatics (Münster)

Doctoral student representative at the Max Planck Institute for Molecular Biomedicine (2017–2021)

Experimental training in the laboratories of Prof. Justin Teissie (2011-2014) and Dr. Elena Gomez-Diaz (2013)

LANGUAGES

Catalan (*native*)

Spanish (*native*)

English (*fluent*)

German (*basic*)

French (*basic*)